

GOOGLE CLOUD PROFESSIONAL CLOUD ARCHITECT

Section 4 -- Operations Cheat Sheet

GCP Study Notes

Study Notes April 2026

Compute Engine

Connect

```
gcloud compute ssh VM --zone=ZONE # Standard SSH
gcloud compute ssh VM --zone=ZONE --tunnel-through-iap # No external IP
gcloud compute connect-to-serial-port VM --zone=ZONE # Boot issues
gcloud compute reset-windows-password VM --zone=ZONE --user=admin # RDP
gcloud compute scp file VM:~/file --zone=ZONE # Copy files
```

VM Lifecycle

```
gcloud compute instances stop|start|reset|suspend|resume|delete VM --zone=ZONE
gcloud compute instances delete VM --zone=ZONE --keep-disks=boot # Keep disk
gcloud compute instances set-machine-type VM --zone=ZONE --machine-type=TYPE # Must be ST
gcloud compute instances move VM --zone=ZONE --destination-zone=ZONE2
gcloud compute instances list --filter="status=RUNNING"
gcloud compute instances describe VM --zone=ZONE
```

Snapshots

```
gcloud compute snapshots create SNAP --source-disk=DISK --source-disk-zone=ZONE
gcloud compute snapshots list
gcloud compute snapshots delete SNAP
gcloud compute disks create DISK --source-snapshot=SNAP --zone=ZONE # Restore

# Scheduled snapshots
gcloud compute resource-policies create snapshot-schedule SCHED \
  --region=REGION --max-retention-days=14 --daily-schedule --start-time=02:00
gcloud compute disks add-resource-policies DISK --resource-policies=SCHED --zone=ZONE
```

Images

```
gcloud compute images create IMG --source-disk=DISK --source-disk-zone=ZONE --family=FAM
gcloud compute images list --no-standard-images # Custom only
gcloud compute instances create VM --image-family=FAM --image-project=PROJ # Latest in fa
gcloud compute images deprecate OLD --state=DEPRECATED --replacement=NEW
```

	Snapshot	Image
For	Backup any disk	Boot disk template
Incremental	Yes	No
Families	No	Yes
VM creation speed	Slower	Faster

:warning: Changing machine type requires VM to be **stopped**. :warning: Deleting a snapshot in an incremental chain is safe – Google migrates data automatically. :warning: Serial console = last resort when SSH/RDP broken.

GKE

Cluster & Inventory

```
gcloud container clusters list
gcloud container clusters get-credentials CLUSTER --zone=ZONE # Connect kubectl
kubectl get nodes -o wide
kubectl get pods -A # All namespaces
kubectl describe pod POD
kubectl logs POD [-c CONTAINER] [--previous] [-f]
kubectl top nodes|pods
```

Node Pools

```
gcloud container node-pools create POOL --cluster=C --zone=Z \
  --machine-type=TYPE --num-nodes=3 --enable-autoscaling --min-nodes=1 --max-nodes=10
gcloud container clusters resize C --node-pool=POOL --num-nodes=5 --zone=Z # Manual scale
gcloud container node-pools update POOL --cluster=C --zone=Z \
  --enable-autoscaling --min-nodes=2 --max-nodes=10
gcloud container node-pools delete POOL --cluster=C --zone=Z
```

Artifact Registry Access

Image URL: LOCATION-docker.pkg.dev/PROJECT/REPO/IMAGE:TAG

- **Default:** Grant `roles/artifactregistry.reader` to node SA
- **Workload Identity** (recommended): Bind K8s SA to GCP SA with reader role
- **ImagePullSecrets:** For cross-project / external registries

Kubernetes Resources

Resource	Use
Deployment	Stateless apps, random pod names, shared PVC
StatefulSet	Stateful apps (DBs), ordered names (pod-0,1,2), individual PVCs

```
kubectl apply -f FILE.yaml
kubectl run POD --image=IMG --restart=Never
kubectl delete pod POD --grace-period=0 --force # Force delete stuck pod
kubectl scale statefulset STS --replicas=N
```

Service types: ClusterIP (internal) | NodePort | LoadBalancer (external)

Autoscaling

Autoscaler	Scales	Trigger
HPA	Pod replicas	CPU/memory/custom metrics
VPA	Pod CPU/memory requests	Historical usage
Cluster Autoscaler	Nodes	Unschedulable (pending) pods

```
kubectl autoscale deployment DEP --min=2 --max=10 --cpu-percent=70
gcloud container clusters update C --zone=Z --enable-autoscaling \
  --min-nodes=2 --max-nodes=10 --node-pool=POOL
```

VPA modes: **Off** (recommend only) | **Initial** (creation only) | **Recreate** | **Auto**

:warning: HPA + VPA on same deployment = conflict. Don't combine on same metric. :warning:
 Pending pods trigger Cluster Autoscaler, not high CPU. :warning: `kubectl get pods -A` = all namespaces (exam shortcut).

Cloud Run

Deploy & Revisions

```
gcloud run deploy SVC --image=IMG --region=R # 100% to new revision
gcloud run deploy SVC --image=IMG --region=R --no-traffic # Deploy without traffic
gcloud run deploy SVC --image=IMG --region=R --no-traffic --tag=canary # Tagged URL
gcloud run revisions list --service=SVC --region=R
gcloud run services describe SVC --region=R --format="value(status.url)"
```

Traffic Splitting & Rollback

```
gcloud run services update-traffic SVC --region=R \
  --to-revisions=REV_OLD=90,REV_NEW=10 # Canary
gcloud run services update-traffic SVC --region=R --to-latest # 100% latest
gcloud run services update-traffic SVC --region=R \
  --to-revisions=PREV_REV=100 # Instant rollback
gcloud run services describe SVC --region=R --format="yaml(status.traffic)"
```

Scaling

Param	Default	Purpose
<code>--min-instances</code>	0	Warm instances (0 = scale to zero)
<code>--max-instances</code>	100	Upper bound
<code>--concurrency</code>	80	Requests per instance
<code>--no-cpu-throttling</code>	off	CPU always on (WebSockets, background)

```
gcloud run services update SVC --region=R --min-instances=2 --max-instances=50
gcloud run services update SVC --region=R --concurrency=50
gcloud run services update SVC --region=R --no-cpu-throttling
```

IAM

```
gcloud run services add-iam-policy-binding SVC --region=R \
  --member="allUsers" --role="roles/run.invoker" # Public
```

:warning: `--no-traffic` + `--tag=canary` = test via tag URL without affecting prod. :warning:

Rollback = just reroute traffic to old revision, no redeployment needed. :warning: `min-`

`instances=0` = cold starts. Set ≥ 1 for latency-sensitive services. :warning: `--no-cpu-throttling` required for WebSockets / background work.

Storage & Databases

Cloud Storage Operations

```
gcloud storage cp|mv|rm|ls gs://BUCKET/PATH
gcloud storage buckets update gs://B --uniform-bucket-level-access # IAM only
gcloud storage buckets update gs://B --versioning
gcloud storage buckets update gs://B --lifecycle-file=lifecycle.json
gcloud storage buckets update gs://B --retention-period=90d
gcloud storage buckets update gs://B --lock-retention-period # IRREVERSIBLE
gcloud storage sign-url gs://B/file --duration=1h --private-key-file=key.json
```

Lifecycle Rules (JSON)

- `SetStorageClass` + `age` condition = auto-transition (Standard -> Nearline -> Coldline -> Archive)
- `Delete` + `age` = auto-delete
- `Delete` + `isLive: false` + `numNewerVersions: N` = prune old versions

Key Storage Roles

Role	Access
<code>objectViewer</code>	Read
<code>objectCreator</code>	Write (no read/delete)
<code>objectAdmin</code>	Full object control
<code>storage.admin</code>	Full bucket + object control

Database Queries

```
gcloud sql connect INST --user=root --database=DB
bq query --use_legacy_sql=false 'SQL'
bq query --dry_run 'SQL' # Cost estimate, no charge
gcloud spanner databases execute-sql DB --instance=I --sql="SQL"
```

Cloud SQL Backups

```
gcloud sql backups create --instance=INST
gcloud sql backups restore ID --restore-instance=INST
gcloud sql instances patch INST --backup-start-time=02:00 --enable-bin-log # PITR (MySQL)
gcloud sql instances clone INST CLONE --point-in-time="2024-06-15T10:30:00Z" # PITR resto
```

Firestore Backup

```
gcloud firestore export gs://BUCKET/path [--collection-ids=coll,col2]
gcloud firestore import gs://BUCKET/path
```

Job Status

```
gcloud dataflow jobs list --region=R [--status=active]
gcloud dataflow jobs drain JOB_ID --region=R    # Graceful stop (finish in-flight)
gcloud dataflow jobs cancel JOB_ID --region=R    # Immediate stop
bq ls -j --max_results=20                        # BQ jobs
bq show -j JOB_ID
```

:warning: `--lock-retention-period` is **irreversible** – cannot shorten or remove. :warning:

Firestore has **no built-in scheduled backups** – use Cloud Scheduler + Cloud Function. :warning:

`bq query --dry_run` = free cost estimate. :warning: Dataflow `drain` vs `cancel`: drain finishes in-flight data, cancel stops immediately. :warning: Storage early deletion fees: Nearline 30d, Coldline 90d, Archive 365d.

Networking

Subnets

```
gcloud compute networks subnets create SUB --network=VPC --region=R --range=10.1.0.0/24
gcloud compute networks subnets create SUB --network=VPC --region=R --range=CIDR \
  --enable-private-ip-google-access --enable-flow-logs \
  --secondary-range=pods=10.8.0.0/14,svcs=10.12.0.0/20    # GKE
gcloud compute networks subnets expand-ip-range SUB --region=R --prefix-length=20
```

Static IPs

```
gcloud compute addresses create IP --region=R                # Regional external
gcloud compute addresses create IP --global                  # Global (for LB)
gcloud compute addresses create IP --region=R --subnet=SUB --addresses=10.0.1.50 # Internal
gcloud compute addresses list
gcloud compute addresses delete IP --region=R
```

Cloud DNS

```
gcloud dns managed-zones create ZONE --dns-name=example.com. --visibility=public
gcloud dns managed-zones create ZONE --dns-name=internal.co. --visibility=private --network
# Records via transactions:
gcloud dns record-sets transaction start --zone=Z
gcloud dns record-sets transaction add "1.2.3.4" --name=www.ex.com. --ttl=300 --type=A --z
gcloud dns record-sets transaction execute --zone=Z
```

Cloud NAT

```
gcloud compute routers create RTR --network=VPC --region=R
gcloud compute routers nats create NAT --router=RTR --region=R \
  --auto-allocate-nat-external-ips --nat-all-subnet-ip-ranges
```

:warning: Subnets can only **expand**, never shrink. One-way operation. :warning: Unused reserved static IPs **incur charges**. :warning: Cloud NAT = **outbound only**. No inbound from internet. :warning: DNS records require transaction workflow: start -> add/remove -> execute. :warning: DNS names must end with a dot (e.g., `example.com.`).

Monitoring & Logging

Alerts

```
gcloud alpha monitoring channels create --type=email --channel-labels=email_address=X
gcloud alpha monitoring policies create --policy-from-file=alert.json
gcloud monitoring uptime create CHK --resource-type=uptime-url --hostname=HOST --path=/hea
```

Key Metrics Quick Reference

Service	Metric prefix
Compute Engine	<code>compute.googleapis.com/instance/cpu/utilization</code>
Cloud SQL	<code>cloudsql.googleapis.com/database/cpu/utilization</code>
GKE	<code>kubernetes.io/container/cpu/request_utilization</code>
Cloud Run	<code>run.googleapis.com/request_count</code>
Load Balancer	<code>loadbalancing.googleapis.com/https/request_count</code>

Log-Based Metrics

```
gcloud logging metrics create NAME --log-filter='severity>=ERROR' --description="DESC"
# Use in alerts: metric.type = "logging.googleapis.com/user/NAME"
```

Log Sinks (Export)

Destination	Use
Cloud Storage	Archival, compliance
BigQuery	SQL analysis
Pub/Sub	Real-time / SIEM

```
gcloud logging sinks create SINK DEST --log-filter='FILTER'
# Grant writer identity access to destination after creation!
gcloud logging sinks describe SINK --format="get(writerIdentity)"
```

Log Filters

```
gcloud logging read 'severity>=ERROR' --limit=20
gcloud logging read 'resource.type="gce_instance"' --limit=20
gcloud logging read 'logName="projects/P/logs/cloudaudit.googleapis.com%2Factivity"'
gcloud logging read 'protoPayload.methodName="v1.compute.instances.delete"'
```

Log Buckets

Bucket	Retention	Configurable
<code>_Required</code>	400 days	No
<code>_Default</code>	30 days	Yes
Custom	1-3650 days	Yes

```
gcloud logging buckets create BUCKET --location=LOC --retention-days=90
gcloud logging sinks update _Default --add-exclusion=name=X,filter='severity=DEBUG'
```

Audit Logs

Type	Default	Cost
Admin Activity	Always on, cannot disable	Free
Data Access	Off (except BigQuery)	Charged
System Event	Always on	Free
Policy Denied	Always on	Free

Ops Agent

```
curl -sSO https://dl.google.com/cloudagents/add-google-cloud-ops-agent-repo.sh
sudo bash add-google-cloud-ops-agent-repo.sh --also-install
# Config: /etc/google-cloud-ops-agent/config.yaml
sudo systemctl restart google-cloud-ops-agent
```

Managed Prometheus

```
gcloud container clusters update C --zone=Z --enable-managed-prometheus
# Then create PodMonitoring CRDs to scrape app metrics
```

:warning: Admin Activity audit logs = **always on, free, cannot be disabled**. :warning: Data Access audit logs = **off by default** (except BigQuery), **cost money**. :warning: Log sink writer identity needs **explicit permissions** on destination. :warning: `_Required` bucket retention (400 days) **cannot be changed**. :warning: Ops Agent replaces **both** legacy Monitoring and Logging agents. :warning: Exclusion filters reduce log storage costs but excluded logs are **gone forever**.